Building an Advice-Only ED Workflow with LLM Collaboration: A Clinician's Engineering Journey

Author: Dr. med. Katharina Jacoby
Date: 30 August 2025
Keywords: emergency medicine, workflow coordination, LLM, safety, governance, Agent Mesh, ICU capacity, troponin 0/1h, ethics, auditability, SPC, MIMIC-IV ED Demo

---

Abstract

Emergency departments (EDs) are coordination bottlenecks where minutes matter and information breaks easily across handovers. This paper narrates my journey as a clinician collaborating with large language models (LLMs) and small ML components to build an **advice-only** workflow assistant for ED operations. The goal was not diagnosis but *coordination*: structured reminders, follow-ups, capacity-aware proposals, and auditable prompts that reduce cognitive load without ceding authority. I describe iterative architectures—from a monolithic learner to a **skills/Agent Mesh** under a safety governor—highlighting failures, debugging tactics, methodological pivots, and the explicit reasons for each change.

Two final pivots are central: (1) replacing **Isolation Forest** anomaly detection with **Statistical Process Control (SPC)** (Shewhart/EWMA) to avoid conflicts with MLP-based gating and to prioritize research integrity and auditability; and (2) abandoning synthetic data in favor of the **MIMIC-IV ED Demo** public dataset after synthetic scenarios proved non-realistic. I conclude with qualitative impacts on acceptance and alert fatigue, constraints (ICU realism, HL7 scope), and a governance contract that keeps the system transparent, bias-aware, and reversible.

---

1. Motivation and Scope

Urban EDs struggle with fragmented orchestration: EMS → triage → diagnostics → specialists → bed/ICU. My objective was to support these **coordination** steps (paging, reminders, pre-populated orders, capacity checks) while enforcing an ethics stance: **no autonomous clinical decisions; advice-only with human approval**. The assistant must be small, auditable, and easy to pause or revert.

---

2. Methods and Architecture

2.1 Initial Baseline (Monolith)

I began with a simple MLP/GRU learner trained on synthetic sequences—initially with a **Mamba-backboned** variant for the sequence block. In retrospect this **GRU + MLP with a Mamba backbone** encouraged over-fitting at our data scale and added tuning burdens with little interpretability gain. We reverted to a plain GRU for temporal dynamics and kept the MLP head simple.

I began with a simple MLP/GRU learner trained on synthetic sequences to propose next administrative/ coordination actions. This validated the feasibility of end-to-end training but surfaced variance and debuggability issues (e.g., distribution gaps, thresholding, calibration). See Figure 1 for early validation behavior.

2.2 Skills + Agent Mesh under a Safety Governor

We pivoted to a **skills network**: small, composable micro-policies (e.g., CT at 60 min follow-up; serial hs-cTnT with strict 0/1-h pairing; consult follow-ups; lingering prompts for patients held >2 h; equipment tracker with QR). A **Coordinator/Governor** mediates proposals, applies cooldowns, deduplication, and justification requirements (e.g., ICU bed request rationale), and logs everything for audit.

Why pivoted: Modularity increased explainability, resilience, and the ability to encode clinical policy without burying logic inside a black-box. It also made it trivial to disable any noisy skill and to study acceptance vs. fatigue per skill.

2.3 Calculators & Signals (Advice-Only)

To anchor transparency, we added phase-2 calculators (qSOFA, MEWS, HEART, Wells, D-dimer with age-adjusted threshold, **Roche hs-cTnT 0/1-h** rule-in/out/observe) and emitted **signals** (e.g., `acs_rulein`) that map to advice-only prompts (page cardiology; confirm serial ECGs). Event logging uses JSONL to enable tailing and audits.

2.4 Capacity-Aware Flows

A fallback agent proposes ED hold or transfer when ICU capacity is constrained; a "joker" identification/ transfer window keeps operations realistic. ICU acceptance rules and negotiation are staged for later realism; current snapshots estimate next-bed ETAs.

2.5 Statistical Process Control (SPC) instead of Isolation Forest

Why pivoted: Isolation Forest sounded attractive for anomaly detection, but in practice it conflicted with the MLP-based gates and produced opaque flags that were hard to reconcile with governance and clinical review. We replaced it with **SPC** (Shewhart/EWMA). SPC's explicit control limits, stable false-alarm behavior, and straightforward audits aligned better with our advice-only stance and reduced cross-model conflict.

How we use SPC: - Shewhart charts for discrete proposal rates (per skill) and page volume. - EWMA for drift of acceptance and fatigue indices. - Clear rules: points beyond limits, 2-of-3 near a boundary → flag for review, not auto-action.

Effect: fewer contradictory triggers with the MLP gate, easier policy reviews, and simpler governance diffs (see Figure 5).

2.6 Data Source Pivot: From Synthetic Generator to MIMIC-IV ED Demo

Why pivoted: Synthetic data created edge-case scenarios that were not clinically realistic and undermined evaluation. We switched to the **MIMIC-IV ED Demo** (public, de-identified) to ground calculators and workflow gates in more representative distributions.

Effect: more credible sanity checks for risk calculators and disposition prompts; clearer failure modes; fewer misleading “wins” from the generator.

2.7 Checklist-Driven Priorities (Phase 1 → 2 → 3)

Drawing directly from the internal checklist, we enforced a strict priority order: - **Phase 1 — Operational Infrastructure (do first):** QR-based equipment tracking (ultrasound, crash carts, wheelchairs, IV pumps), SOP quick-access with flowchart/checklists (~30 protocols), and **lingering patient monitoring** (last-seen timers, comfort basics, handoff alerts). - **Phase 2 — Clinical Integration (after Phase 1 works):** HL7v2 results processing with troponin delta logic; auto-calculation of HEART/GRACE/Marburg, qSOFA/MEWS/SOFA; context tracking (meds, allergies, pregnancy, prior results) with **advice-only** prompts. - **Phase 3 — Advanced Features (last):** workflow optimization (queues/bed assignment/consult coordination), handoff coordination, and lightweight predictive components for deterioration/LOS/capacity.

This order prevented the “academic trap,” kept focus on 15-minute operational problems, and made acceptance our primary success metric.

2.8 LLM Collaboration Methodology ("Vibe-Coding" Under Contract)

We used a **contract-driven collaboration** with LLMs. The contract enforces safety, provenance, and auditability. When uncertain, the LLM **must ask**—no guessing.

Contract Overview

Invariants (MUST NOT CHANGE/REMOVE) - `WorkflowState` unchanged, defined before use; exposes `.feature_dict()`, `.update_state_from_event(...)`, `.apply_event_log(...)`; timers/backlogs/alerts intact. - `TinyCritics` uses `WorkflowState.feature_dict()`; cold-start safe (no transform before fit). - `CONFIG` bootstrap with defaults; flags `RUN_UI=False`, `RUN_PIPELINE=False`. Paths exist and remain CSV/dir semantics: `EQUIPMENT_STATUS_PATH`, `EQUIPMENT_MOVES_LOG_PATH`, `SOP_REGISTRY_PATH`, `QR_OUTPUT_DIR`, `EVENT_LOG_PATH`. - Tracker core first-class (`tracker_core.TrackerService` + `tracker_ui.run_ui(...)` or inlined equivalents). No import-time side effects. - SOP auto-pull present & offline-safe: `refresh_sop_registry(CONFIG, base_url)`; lazy imports; returns summary on failure. - QR: generate + scan-to-update; manual payload fallback must still update moves when decode libs absent. - Overdue/alerts: equipment overdue (by last_seen) + lingering patient indicator (by since_vitals_min) with adjustable thresholds in UI. - Notebook metadata has kernelspec: `python3`.

No-hallucinations / verification rules (MANDATORY) - No claims without evidence: do **not** state "implemented," "wired," or "working" unless: - Code cell(s) are added/modified in this notebook or a file write is performed, **and** - A local check/smoke assert for that feature runs in the same response (or in the Sanity Pack) and passes. - Provenance required: when referring to features from older notebooks or sidecars, name the file(s) and cell/module copied from or superseded. - State uncertainty explicitly: if an external dependency, network, or environment capability is required, say so and describe fallback behavior. Do not imply it succeeded unless verified. - No virtual resources: don't reference files/URLs/data that weren't created in this run or already present. If missing, create them or mark as TODO—don't claim they exist. - Exact diffs or full cells: any modification must include a line-precise diff **or** the full replacement cell/module content. - No silent renames/moves: if a symbol/path changes, declare the change and update all call sites in the same patch; otherwise, don't change it. - Ask before destructive edits: never delete code/data without an explicit instruction.

Allowed changes - New features only behind guards: `if RUN_PIPELINE:` (Phase-2/3) and/or `if RUN_UI:`. - Append after core definitions; do not reorder `WorkflowState`/`TinyCritics`. - Optional deps (`requests`, `bs4`, `Pillow`, `pyzbar`) must be lazy-imported inside guarded code. - You may extend `.feature_dict()` by adding keys (no renames/removals).

Forbidden changes - No placeholders for `WorkflowState`. No renames of core classes/functions/paths. - No top-level execution (no jobs at import; no undefined symbol instantiation). - No order fragility (no references before definition). - Do not touch PDFs/data except via tracker/SOP services.

Delivery checklist (MUST PASS)

```
# 1. Set:
CONFIG["RUN_UI"] = True; CONFIG["RUN_PIPELINE"] = True

# 2–6. TinyCritics cold-start
s = WorkflowState(role="nurse"); getattr(s, "touch_now", lambda *_: None)
(pd.Timestamp.utcnow())
tc = TinyCritics(); p, b, u = tc.score(s, [
    {"id":"reassess_vitals","label":"Reassess vitals"},
    {"id":"order_ecg","label":"Order ECG"}
])
assert len(p) == 2 and (0 <= p).all() and (p <= 1).all()

# 7–15. Tracker core basic IO
from pathlib import Path
from tracker_core import TrackerService, QRService, EquipmentRepository,
MovesLogRepository, SOPRegistry

t = TrackerService.from_config(CONFIG)
_ = t.equipment_status(); t.log_move("pump-001","A1","B2"); assert
Path(CONFIG["EQUIPMENT_MOVES_LOG_PATH"]).exists()
q = QRService(CONFIG["QR_OUTPUT_DIR"]).make("poctest"); assert isinstance(q,
str) and len(q) > 0
sop = SOPRegistry(CONFIG["SOP_REGISTRY_PATH"]).read(); assert sop is not None
print("SMOKE_OK")

# 16–20. Surfaces and metadata
```

```
# SOP auto-pull surface present; offline or missing libs returns a non-throwing summary.
# QR scan fallback present: manual payload (e.g., id=pump-001) updates location and appends to moves log if decode fails.
# Overdue/alerts visible in UI with adjustable thresholds (minutes).
# Notebook metadata contains: {"kernelspec":
{"name":"python3","display_name":"Python 3","language":"python"}}
```

Reviewer Loop (A → B → C, then back) 1) **Implement & question** with LLM **A** under the contract; when A is uncertain, it must ask.
2) **Human validation**: run smoke tests, adjust, and answer A's domain questions.
3) **Critical review** with LLM **B** (detect overblown code, placeholders, hallucinations, brittle design).
4) Iterate with **B** until next steps are clear or code is implemented; validate again.
5) **Independent review** with LLM **C**; validate again.
6) Open a new session; hand off the updated notebook + contract back to **A/B** for divergence testing; choose the stronger direction and proceed.

---

3. Debugging the Journey — Fails → Fixes → Lessons

3.1 Code/Environment Breakers

- Missing or malformed state machines and config bootstrap; unterminated Python strings.
- Recursive default overrides; PyTorch `DataLoader` sampler/shuffle conflicts.
- CUDA/lib issues blocking confusion-matrix generation.
- Kaggle/Papermill notebook metadata missing (`kernelspec`), breaking auto-runs; whole notebook JSON pasted into a single code cell (!).

Fixes: - Reconstructed notebooks with valid cells, injected `kernelspec` metadata, and added a **preflight** cell that halts on invariants failures. - Path guards (`/kaggle/working`), directory creation before writes, and CONFIG bootstrapping (frozen defaults, overlay allowed but no silent mutation). - File-backed fallbacks for fragile imports; synthetic data generator made portable; complaint distributions corrected; negative ETA clamped; gate threshold calibrated.

Lessons: - Always ship a smoke pack: tiny critics, workflow state round-trip, QR make/scan, SOP registry, and a pass/fail domain audit. - Freeze contracts (what functions, paths, keys must exist) and only append new code behind `RUN_*` guards.

3.2 Flow & Policy Misconceptions

- Assumed CT slower at night; site reality disagreed. We normalized CT follow-up to a fixed 60 minutes regardless of shift.
- ICU triggers were too conservative (no requests firing while boarding surged). We instrumented preconditions and retry loops.

3.3 Alert Fatigue and Acceptance

Early runs either produced few suggestions (ineffective) or repeated nags (fatiguing). Anti-spam gates (cooldown, snooze-after-accept, dedupe) and justification enrichment improved acceptance. Figure 2 summarizes acceptance vs. fatigue across iterations.

---

4. Architecture Evolution and Rationale

We advanced through four stages:

1. **Baseline MLP** — debuggable but limited temporal reasoning.
2. **GRU sequence model** — captured temporal patterns; validation AUC improved but test variance highlighted synthetic distribution gaps (Figure 1).
3. **Agent Mesh + Governor** — modular, explainable skills; safety gates and audit trails; better control of alert fatigue.
4. **Phase-2 calculators + gates** — transparent, advice-only outputs; easy to verify and to tune with clinicians in the loop.

A concise table is provided in the repository (Architecture Evolution Summary CSV) and linked from this paper.

---

5. Evaluation

5.1 Model Behavior (Early Training)

Validation AUCs climbed across epochs in the GRU baseline, while a small test set showed high variance—underscoring the need for repeated CV and calibration. *Figure 1.*

Figure 1 — Validation AUC by Epoch (Day 7 GRU baseline)

5.2 Operational Impact (Advice-Only Mesh)

Later runs with the mesh and governor recorded **acceptance** ~0.45 and **fatigue** ~0.55 (proxy), after anti-spam measures and clearer justifications; early runs showed acceptance near 0.0 with fatigue ~1.0 (Figure 2).

Figure 2 — Clinician Acceptance vs. Alert Fatigue

5.3 Disposition/Flow Snapshot

On a small test snapshot, we observed high boarding with ICU requests under-firing. Instrumentation and retry logic were added accordingly. *Figure 3.*

Figure 3 — ED Disposition / Flow Rates (Day 7 Test Snapshot)

5.4 Architecture & Data Pivots (Visual Summary)

The timeline shows the design path: Baseline MLP → GRU+MLP (Mamba attempted) → Agent Mesh + Governor → Calculators + Gates → **SPC Monitoring** replacing Isolation Forest, with a **data pivot to MIMIC-IV ED Demo** before SPC adoption. *Figure 4.*

Figure 4 — Architecture & Data Pivots Timeline

5.5 SPC Pivot Rationale (Qualitative Indices)

Qualitative indices (for illustration) comparing Isolation Forest vs SPC along conflict with MLP gates, auditability, and operational stability. *Figure 5.*

Figure 5 — SPC Pivot Rationale — Qualitative Indices

6. Governance, Safety, and Bias Mitigation

- **Advice-only** stance; human approval required for orders/transfers.
- **Audit everything** (JSONL): who/when/why for proposals and approvals; reproducible thresholds.
- **Feature ethics:** excluded age/sex from decision logic; risk scores can display these but are not drivers.
- **No network calls** or external dependencies during protected phases; privacy-respecting HL7 parsers and local calculators.
- **Invariants contract:** protected functions, paths, and semantics; append-only edits with proof of preflight and smoke tests.

6.1 AI-Generated Code & Authorship Disclosure

Portions of this prototype (including scaffolding, utilities, and small UI/helpers) were **generated with an AI assistant** and then **reviewed, edited, and tested** by **Dr. med. Katharina Jacoby**. All clinical content, architectural decisions, and acceptance criteria are authored and owned by the clinician-author. AI outputs were used to speed implementation, but **no clinical decisions are automated** and **all code paths are covered by smoke tests** and governance guards.

Provenance & integrity rules followed: - Adhered to a modification **contract** with protected surfaces, guards (`RUN_UI`, `RUN_PIPELINE`), and explicit preflight/smoke checks. - Cited provenance when reusing prior cells/snippets; avoided hallucinated APIs; provided diffs for appended cells only. - Ensured **license safety**: accepted MIT/BSD/Apache-2.0 snippets only; avoided copyleft/unknown-license code.

6.2 Reproducibility & Provenance Checklist (for reviewers)

- **Prompt/Session Logs:** Kept minimal logs of LLM prompts relevant to code generation and architectural choices.
- **Model/Tool Versions:** Recorded scikit-learn/PyTorch versions; pinned where possible.
- **Seeds & Data:** Stored seeds; moved from synthetic generator → **MIMIC-IV ED Demo** for sanity-checked distributions.
- **Smoke Tests:** Preflight cell; hs-cTnT 0/1-h rule-in/out/no-pair cases; event-log tail check.

- **Artifacts:** Saved figures, CSV summaries, and JSONL logs for audits.
- **Human Review:** Every AI-generated snippet was read, edited as needed, and executed under guards before inclusion.

6.3 Scientific Integrity & Transparency

- **Preregistration & protocol:** Primary/secondary endpoints, metrics (AUPRC, AUROC, ECE, FP/hour at recall τ), subgroup plan, and rollback rules are specified **before** training; any evaluation begins in **shadow mode**.
- **Data governance & compliance:** Uses **MIMIC-IV ED Demo** (public, de-identified) under its data-use terms; dataset version and access date recorded; no PHI enters the advice-only system; data access roles are documented.
- **Reproducibility bundle:** Release code, seeds, frozen environment (`env.yml`/Docker), commit hash, and scripts to regenerate all figures/tables and the trained MLP bundle (one-command `make reproduce`).
- **Baselines & ablations:** Include logistic-regression and simple rule-based baselines; run ablations per feature group and SPC-threshold sensitivity.
- **Fairness & subgroup calibration:** Pre-specify slices (e.g., site, time-of-day, arrival mode, age band, sex-if-reported) and report calibration + error rates with 95% CIs. Age/sex are not features but are used for reporting.
- **Governance & incident response:** SPC (Shewhart/EWMA) with clear review triggers, alert budgets, a **kill-switch/rollback**, and named sign-off roles.
- **Security & privacy:** SBOM/dependency scan; no secrets in repo; access logging; threat model for data exfiltration, prompt-injection via notes, and QR misuse.
- **Licensing & citation:** Code license and data availability statement; provide **CITATION.cff** and a DOI plan (e.g., Zenodo).
- **Compute & sustainability:** Report hardware, total train time, and approximate energy/cost.
- **Authorship & AI-assist provenance:** CRediT roles; conflicts of interest = none; AI-generated code disclosed and human-reviewed (Section 6.1).

7. Discussion

Moving from a monolith to a skills mesh shifted effort from chasing metrics to **making policies explicit** and measurable. The mesh enabled co-evolution with ED needs: any skill can be tuned or retired with isolated blast radius. Acceptance became a first-class metric, not an afterthought. Transparent calculators (hs-cTnT 0/1-h, D-dimer, risk scores) grounded the LLM collaboration: the LLM orchestrates, explains, and documents rather than "decides."

---

8. Limitations

- Synthetic data and small test sets limit external validity. (Although we pivoted to MIMIC-IV ED Demo, larger-scale evaluation is pending.)
- ICU capacity remains a demo snapshot; no live bed API.
- HL7 ingestion covers targeted lab OBX segments; a fuller parser is future work.
- Confusion matrices for sequence models were deferred due to CUDA constraints in this environment.

- **No independent evaluation by a human data scientist yet;** current assessments are clinician-led with an LLM A/B/C reviewer loop. A formal DS review of labeling, leakage checks, calibration, and code quality is planned.

---

9. Future Work

1. **Independent data-scientist evaluation:** hand off the notebook + artifacts for external replication, leakage audit, calibration check, and code review; integrate findings.
2. Scenario-card runner for canonical ED presentations (STEMI, neuro deficit, trauma, early sepsis) with top-k proposals and minutes-saved estimates.
3. Pager-load by hour with adjustable OPS budgets and freshness decay tuning.
4. Bias audits using counterfactual toggles; publish fairness metrics by subgroup.
5. Stronger ICU realism (negotiation, acceptance rules) and capacity staleness modeling.
6. Exportable artifact pack (zip) with KPIs, calibration plots, and JSONL logs for review boards.

---

11. Outlook — Clash-Free MLP Classifier Integration

Goal (one label)
Start with a single coordination KPI label (e.g., `y_ecg_repeat_60m`, `y_trop_pair_0_1h`, or `y_result_to_page_10m`). Keep it advice-only and evaluate via minutes-saved/compliance.

Contract-safe interfaces
Preserve the public API; add a new bundle path; guard training with `RUN_PIPELINE`; write minimal `ml_score` events without changing existing schemas.

Feature schema (append-only)
Extend `feature_dict()` with timers, context flags, calculator outputs, ops snapshots, and missingness bits—never rename or remove.

Labeling (no leakage)
Derive silver labels strictly from future-of-t0 events; enforce time cutoffs so features never peek at outcomes.

Training recipe (lean + reproducible)
MLP 64→32 (ReLU, dropout 0.1) with class weights; LogisticRegression baseline; 5× CV; report AUPRC/AUROC/**ECE**; isotonic calibration; save bundle, features, calibration, report; set thresholds by recall floor + FP/hour budget.

Integration points (zero clash)
Add **MLPGate** that emits **advice-only** suggestions at `yhat≥τ`; feed positive-rate to **SPC** (Shewhart/EWMA) for drift monitoring; no auto-tuning. UI text: “suggests X (p=0.xx).”

Smoke & sanity checks
Bundle loads and bounds hold; feature parity vs `features.json`; ECE ≤ target; SPC chart wired; KPI summary printed.

CONFIG additions (append-only)
Add `CONFIG["MLP"]` with `TARGET`, `FEATURES_VERSION`, `BUNDLE_PATH`, `THRESHOLD`, `CALIBRATION`, `SEED`. Do not repurpose existing keys.

Roadmap to larger dataset
Train and calibrate on **MIMIC-IV ED Demo** now; then migrate to the larger dataset, compare distribution shift, re-tune thresholds under the same FP/hour budget; SPC watches post-swap.

Do-next checklist
Choose the label → append features → implement guarded trainer → add MLPGate + SPC watcher → add preflight cell (3 rows) → ship artifacts and report.

---

10. Acknowledgments

This work grew from a clinician-led, ethics-first stance on AI in acute care. Collaboration with LLMs helped keep the focus on clarity, auditability, and equitable workflows.

Appendix A — Architecture Evolution (CSV)

A concise CSV summarizing stages, reasons for pivots, and observed impact is available alongside this paper.

- `architecture_evolution_summary.csv`

Appendix B — Reproducibility & Repository Checklist

- **Reproduce in one command:** `make reproduce` (or `python scripts/reproduce.py`).
- **Environment:** `env.yml` or `Dockerfile` with frozen versions; print versions in paper.
- **Artifacts (`ARTIFACTS/`):** `mlp_v1.joblib`, `features.json`, `calibration.json`, `training_report.json`, `spc_config.json`, and a sample **event log** (`events_sample.jsonl`).
- **Tests:** unit tests for calculators (hs-cTnT 0/1-h edge cases), feature parity, and preflight/smoke.
- **Security:** `SECURITY.md` with threat model + SBOM link; no secrets committed.
- **Governance:** `GOVERNANCE.md` (SPC rules, alert budgets, rollback procedure).
- **Citation:** `CITATION.cff` and DOI plan (Zenodo or similar).

Figure Links (download)

- Figure 1 — Validation AUC by Epoch: `fig1_val_auc.png`

- Figure 2 — Acceptance vs Fatigue: `fig2_acceptance_fatigue.png`
- Figure 3 — ED Flow Rates Snapshot: `fig3_flow_rates.png`
- Figure 4 — Architecture & Data Pivots Timeline: `fig4_architecture_timeline.png`
- Figure 5 — SPC Pivot Rationale — Qualitative Indices: `fig5_spc_pivot_indices.png`